Cache Me if You Can: Repairing the KV State after Compression

Anonymous Authors[1]

Abstract

As large language models (LLMs) support longer contexts and more capable agentic sessions, what matters in the accumulated context can change with each new query, tool output, test result, or browser observation. At the same time, inference still depends on a limited active key-value (KV) cache, and many existing KV-cache compression methods decide which past tokens stay active from the current prefix. As a result, tokens that matter in later turns may already have been evicted or compressed away in earlier compression decisions. We study post-compression KV cache repair, a runtime mechanism that restores previously evicted tokens to the active cache. REPAIRKV uses the interval between turns to reevaluate which tokens are important from evicted KV rows stored in a slower memory tier, then promotes a small subset before decoding resumes. Controlled experiments show that this recovery persists when relevance changes and across different initial eviction policies (compression styles), and a preliminary diagnostic on open-source repositories shows the same pattern; on Qwen2.5-7B-Instruct at 32K context, REPAIRKV achieves 91.7% retrieval on a four-query needle-in-a-haystack task versus 20.8% for the matched no-repair baseline at the same active-cache budget, with only 96 promoted tokens, while the GPU-resident repair operation costs ∼110 ms p95 against ∼2.13 s for full-prefix prefill at 32K (a 10–19× ratio).

1. Introduction

Motivation. Long-context inference increasingly depends on KV-cache compression and offload. During decoding, a transformer stores key-value (KV) activations for prior tokens so later tokens can attend to them. The model weights are fixed once loaded, but the KV cache grows with the number of tokens kept for each live sequence; at long context lengths, this sequence-dependent state can dominate available GPU memory and increase inference-time attention and cache-management costs. KV-cache compression and offload reduce this pressure by deciding which historical state remains active on GPU and which state is evicted or moved to a slower tier. That decision is usually made with information available at the current turn, so it is often treated as a one-way operation: evict now, decode later from the retained active set.

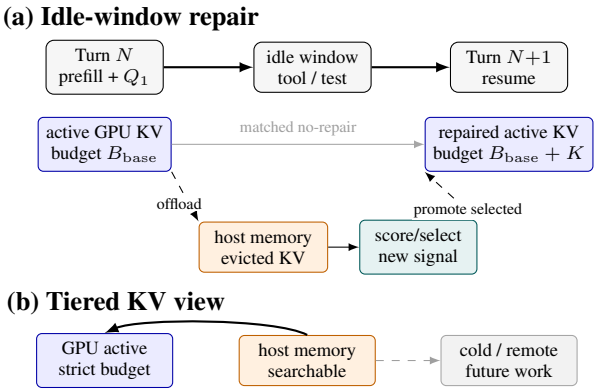

Figure 1. REPAIRKV System Overview: paused-request KV state repair. A paused request offloads evicted KV to host memory, scores and promotes a budgeted subset conditioned on the chosen signal s, and exposes a conceptual tiered-memory interface that production engines would materialize through page or block allocators.

This static view misses two facts about multi-turn inference. First, attention relevance is query- and context-dependent: later turns, retrieved documents, tool outputs, test failures, or file changes can make previously low-priority context newly important. Query-aware and fine-grained KV retrieval systems show that both token criticality and attention-distribution patterns can change with the active query (Tang et al., 2024; Wang et al., 2025). Second, agentic workflows often contain non-decoding intervals while the model waits for external results. Recent systems make this operationally visible: Continuum schedules multi-turn agent requests around tool-call gaps and KV-cache lifetimes, while AgentServe targets local agent serving under tight GPU

[1]Anonymous Institution, Anonymous City, Anonymous Region, Anonymous Country. Correspondence to: Anonymous Author <anon.email@domain.com>.

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memory budgets (Li et al., 2026; Zhang et al., 2026). These systems show that turn boundaries, cache lifetime, and tool-call waits are now explicit systems objects, but they preserve or schedule cached state rather than repairing an already-compressed active cache using the next-turn signal.

Coding-agent measurements reinforce why this interval is worth studying: OS-level execution, including tool calls and setup, can account for 56–74% of task latency (Zheng et al., 2026). Furthermore, modern production serving stacks such as vLLM (Kwon et al., 2023) now deploy prefix caching by default, but re-deriving evicted KV through a fresh prefill pass remains the fallback on cache miss, eviction, or paused-request discard. The cost is severe: prior work reports recompute consuming as much as 99% of multi-turn prefill time when historical KV is repeatedly recomputed (Gao et al., 2024), and 37% of end-to-end execution time when paused-request KV is discarded in agentic workloads (Abhyankar et al., 2024), which motivates tiered KV systems that trade recompute for offloaded reuse (Liu et al., 2025; Dzikanyanga et al., 2026). Together, these factors motivate repairing the active KV state before the next decode.

This paper asks whether a runtime can improve the cache state seen by the next decode after compression has already run. Compression is not only a storage-saving step: it decides which past tokens remain immediately available. At a pause boundary, the runtime may hold a compressed active cache plus a host-memory (CPU DRAM) tier of evicted KV rows, along with a newly available signal s. In shared serving this interval is a scheduler tradeoff; in the single-session or dedicated setting we target, it is often an idle window for the paused request. The central question is whether, given a budget K and any available signal s, the runtime can promote evicted KV rows to improve next-decode quality under the same resumed active-cache budget.

Setting. Recent analysis shows that KV compression can sharply degrade instructions and facts that later become important (Chen et al., 2025). The missing evaluation question is not simply whether compression loses information, or whether a serving system can preserve state across a pause. It is whether future-turn information can improve an already-compressed cache under the same resumed active-cache budget. A useful benchmark for this question needs explicit turn boundaries, a hidden future query, shared pre-turn history, and no-repair controls matched on resumed-cache budget $B_{\text{base}} + K$. In our experiments, the future relevance signal is an explicit second-turn (Q_2) retrieval query so that the repair effect is measurable, while the intended workflow claim is broader than query answering.

Contributions. REPAIRKV is an inference-time adaptation operator for tiered KV state, conditioned on the newly available next-turn signal. The adaptation is non-parametric: model weights, prompts, and decoding settings are fixed, while the active cache state is revised under a resource budget. After turn N, the runtime compresses the evictable context portion to an active set C_{base} with $|C_{\text{base}}| = B_{\text{base}}$ and retains evicted KV in host memory as W_N. When the turn-$(N{+}1)$ signal arrives, REPAIRKV scores that store and promotes a subset $S_K(Q_2) \subseteq W_N$, $|S_K| \leq K$, back into the active GPU cache before decoding resumes. Unlike pause/resume serving or within-turn query-aware compression, REPAIRKV changes a paused, already-compressed cache without rerunning full-prefix selection. We contribute:

- a matched resumed active-cache-budget protocol for the post-compression, pre-resume lifecycle stage, isolating cache repair from simply keeping more active tokens.
- calibrated evidence on pooled split-query multi-query needle-in-a-haystack (MQ-NIAH) 4Q/6Q/8Q evaluations, a five-turn revisit diagnostic, three first-stage eviction policies, a Scissorhands-style persistence stress test, and preliminary real-repository diagnostics.
- a repair prototype with transfer/promotion feasibility results and a controlled GPU runtime probe for chunked scan, top-K selection, and KV movement.

Within this controlled scope, the results suggest that long-context agents may need between-turn KV state updates in addition to pre-generation pruning.

2. Related Work

Recent KV-cache compression and retrieval methods such as SnapKV, StreamingLLM, H_2O, and QUEST define the current frontier for memory-efficient long-context inference. SnapKV, StreamingLLM, and H_2O decide which past tokens remain active under a fixed budget, while query-aware methods such as QUEST emphasize that token criticality depends on the active query (Li et al., 2024a; Xiao et al., 2024; Zhang et al., 2023; Tang et al., 2024). Long-context and KV-lifecycle benchmarks stress contexts from ordinary long documents to 100K-token regimes, showing that application-visible context length does not remove the need for memory-management mechanisms (Bai et al., 2024; Zhang et al., 2024; Hsieh et al., 2024; Li et al., 2025). SCBench makes the KV-cache lifecycle itself a benchmark target, and ShadowKV shows that sparse KV reconstruction and offload can improve long-context serving throughput (Li et al., 2025; Sun et al., 2025). Fine-grained KV retrieval systems dynamically fetch query-relevant rows from larger stores during active inference (Tang et al., 2024; Wang et al., 2025; Qi et al., 2026); attention analyses likewise show that token importance can be query-dependent or recurrent rather than fixed once compressed (Wu et al., 2024; Zhang et al., 2025). These works motivate dynamic KV state, but they mostly make the cache dynamic during active inference: token-eviction methods (SnapKV, H_2O, Scissorhands) discard

rows destructively, while sparse-attention/retrieval methods (Quest, FIER, ShadowKV) keep the full cache and skip or fetch rows during attention. REPAIRKV occupies a third position: rows leave the active GPU cache but persist in host memory, allowing the active set to be revised between turns without rerunning full-prefix selection.

Serving systems provide the runtime context for this question. PagedAttention/vLLM improves GPU KV allocation and block management during serving (Kwon et al., 2023), while InferCept preserves paused requests to avoid recomputing prefixes when augmented generation resumes (Abhyankar et al., 2024). Agent and tool-use benchmarks make external actions part of the workload, including API calls, web interaction, repository edits, and interactive tasks (Qin et al., 2024; Zhou et al., 2024; Jimenez et al., 2024; Liu et al., 2024). Agent-serving and KV-tiering systems reason about resume prefills, offloading, cache reuse, and tiered storage across GPU, host memory, storage, and network tiers (Li et al., 2026; Zhang et al., 2026; Liu et al., 2025; Dzikanyanga et al., 2026; Gao et al., 2024). These systems determine where KV state lives and when it can be reused. InferCept and CachedAttention show how preserved or preloaded KV can avoid recompute; REPAIRKV instead studies which preserved rows should re-enter the active cache when a later turn reveals a new relevance signal.

3. Method

Overview. Each compression step can only choose the active KV set with the information available at that point. Before a repaired cache is used for decoding, any available signal about the upcoming computation can guide which offloaded KV rows should re-enter the active cache. We study a minimal controlled repair mechanism: promote selected offloaded KV rows back into the active cache. Figure 1 summarizes the system overview. At the pause boundary, the abstraction has three objects:

- C_{base}: active evictable KV, with $|C_{\text{base}}| = B_{\text{base}}$ (base active-cache budget);
- W_N: offloaded evicted KV, hidden from decoding unless promoted;
- s: the signal at the pause boundary, i.e. any tokens the runtime uses to score evicted KV, such as a new user query, a tool result, the model's recent generation, or other tokens that signal upcoming attention or relevance.

The repair map chooses $S_K(s) \subseteq W_N$ with $|S_K| \leq K$, where K is the restore budget (the number of evicted KV rows promoted back into the active cache), and resumes from $C_{\text{repair}} = C_{\text{base}} \cup S_K$ at resumed active-cache budget $|C_{\text{repair}}| \leq B_{\text{base}} + K$.

Lifecycle position. Active-attention retrieval methods such as Quest, FIER, and ShadowKV (Tang et al., 2024; Wang et al., 2025; Sun et al., 2025) reselect rows during decoding from a retained or indexed full cache; pause-aware preservation systems (InferCept, Continuum, CachedAttention (Abhyankar et al., 2024; Li et al., 2026; Gao et al., 2024)) keep KV across pause boundaries. REPAIRKV instead applies query-aware scoring at the post-compression, pre-resume boundary – once per pause, against an evicted store created by the compressor itself, with no persistent index. Appendix C.1 attributes the lift to a lifecycle-slot contribution and a separate burst-expansion contribution.

Two-turn protocol. The protocol proceeds in two turns. Two turns is the cleanest minimal case for testing repair under a single relevance shift; the five-turn revisit diagnostic (Figure 3) confirms the effect persists across repeated shifts. At turn N (the first of the two turns), the model receives a long distractor document followed by prompt Q_1, and generates the turn-N answer tail from the full uncompressed cache. The runtime then compresses the distractor document to base budget B_{base} on GPU and offloads the evicted rows to host memory as W_N, while Q_1, the answer tail, and (later) Q_2 are preserved exactly. During the subsequent tool-call idle window, the turn-$(N{+}1)$ prompt Q_2 becomes known. The distractor document is never re-presented at turn $N{+}1$: REPAIRKV scores W_N against the active post-Q_1 cache and the Q_2 signal, promotes a K-budgeted subset S_K back to GPU before Q_2 decoding, and the model answers Q_2 from C_{repair}. Note that we set aside full-prefill recompute baselines from the primary comparison, since re-deriving KV from token IDs after Q_2 is revealed varies the prefill term rather than the resumed active-cache budget. Compute-matched recompute baselines are deferred to future work (Section 6). This setup isolates the repair step from broader serving-system effects by evaluating one paused request on one GPU under three controlled conditions: a single host-memory tier, the full Q_2 signal known before resumed decoding, and a pre-materialized evicted-KV store. Scheduling, remote tiering, and cache contention are separate systems questions, discussed as future directions (Section 6).

Repair operation. Intuitively, REPAIRKV asks which evicted KV rows the chosen signal s (Q_2 in our prototype) would attend to most and promotes a small budgeted subset back into the active cache before decoding resumes, packed into local neighborhoods around the top-ranked anchors. Algorithm 1 states the framework: turn-by-turn decoding and compression are interleaved with an idle-window step that scores, ranks, and selects a K-budgeted subset of evicted rows to promote before the next decode. Two repair-specific choices, SCORE and SELECT, fully specify the prototype; in our prototype, $\mathcal{C} = C_{\text{base}}$ and $\mathcal{W} = W_N$.

Algorithm 1 General REPAIRKV Execution Framework

Input: turn signals $\{s_N\}_{N=1}^{T}$, eviction policy $\mathcal{P}$, base budget B_{base}, restore budget K.
Initialize: active cache $\mathcal{C}$, evicted store $\mathcal{W} \leftarrow \emptyset$.
for $N = 1$ to T **do**
  $\text{ans}_N \leftarrow \text{DECODE}(\mathcal{C}, s_N)$ // Decoding
  $\mathcal{C}, \mathcal{W} \leftarrow \mathcal{P}(\mathcal{C}, \mathcal{W}, B_{\text{base}})$ // Compression
  obtain s_{N+1} // Idle window begins
  for $i \in \mathcal{W}$ **do**
    $a_i \leftarrow \text{SCORE}(i \mid \mathcal{C}, s_{N+1})$ // Scoring
  end for
  rank $\mathcal{W}$ by a_i // Ranking
  $S_K \leftarrow \text{SELECT}(\mathcal{W}, K), |S_K| \le K$ // Selection
  $\mathcal{C} \leftarrow \mathcal{C} \cup S_K$; $\mathcal{W} \leftarrow \mathcal{W} \setminus S_K$ // Repair
end for

For SCORE, post-RoPE Q_2 query tensors $\{q_{2,t}^{(\ell,h)}\}$ are matched against keys in the active and offloaded sets via

$$A^{(\ell,h)} = \text{softmax}\left(\frac{q_{2,:}^{(\ell,h)} [K_{C_{\text{base}}}, K_{W_N}]^\top}{\sqrt{d_k}}\right), \quad (1)$$

restricted to columns in W_N, then aggregated as $a_i = \text{mean}_{\ell,h} \max_t A_{t,i}^{(\ell,h)}$. Because the softmax normalizes over the active and offloaded keys, a_i is the attention mass row i would receive given C_{base} rather than a raw inner product. Ties break by the per-token score from the turn-N first-stage compression, then by ascending position.

For SELECT, each top-ranked anchor i is expanded into a local burst $\{i-L, \ldots, i+R\} \cap (W_N \setminus S_K)$ under a strict $|S_K| \le K$ budget; the right-biased $(L, R) = (2, 20)$ reflects the query supplying the anchor while answer-bearing values trail it. If anchor bursts do not exhaust K, remaining slots backfill from single ranked rows. Burst widths are fixed in this prototype.

The current prototype scores token rows, but the systems abstraction is promotion over cached units; a systems implementation would batch these promotions into exposed blocks or pages.

Matched-budget evaluation. All main claims compare REPAIRKV to a no-repair baseline at the same resumed active-cache budget. A matched resumed-cache budget means that Q_2 decoding sees the same number of active evictable-context tokens in addition to the identical preserved Q_1 prompt and answer tail. It does not mean that the runtime stores the same amount of offloaded KV state. Let B_{match} denote the no-repair condition that applies the base eviction policy with active evictable-context budget $B_{\text{base}}+K$ while preserving the same Q_1 prompt and answer tail. REPAIRKV reaches the same $B_{\text{base}} + K$ active-cache budget by starting from B_{base} retained rows and promoting up to K rows from the host-memory store in this prototype. Extra store bytes, scorer time, and transfer time are not part of the matched active-cache budget; they are reported separately as service costs. The main restore-budget frontier reports raw answer score so readers can compare absolute quality across task difficulty levels. For effect-size comparisons, we also report score gain over matched no-repair,

$$\Delta_{\text{repair}} = \text{Score}(\text{RepairKV}) - \text{Score}(B_{\text{match}}) \quad (2)$$

where Score is the mean fraction of held-out Q_2 targets recovered correctly per example, averaged over the fixed evaluation set shared across all conditions for that setting. This retrieval score measures target recall, while unconstrained answer quality is outside the controlled benchmark. Stricter exact-set and precision-sensitive scoring serve as confirmatory checks. Thus the protocol tests whether a newly revealed turn-$(N+1)$ signal can improve a compressed cache relative to matched no-repair eviction under the same active-cache budget.

4. Experimental Setup

Benchmark family. The main evaluation uses Qwen2.5-7B-Instruct (Qwen, 2024) at 32K rendered context on a single RTX PRO 6000 Blackwell GPU, with context-only SnapKV-style compression (applied only to the evictable document context, with the Q_1 prompt and answer tail preserved) as the initial post-Q_1 compressor applied before repair (Li et al., 2024a). The task family is a split-query version of multi-query needle-in-a-haystack (MQ-NIAH) derived from RULER (Hsieh et al., 2024). Each needle binds a key to a value inside a long distractor context, and the model must return the values for the keys requested in the current turn. The 2Q, 4Q, 6Q, and 8Q variants use the same underlying task template. The name denotes how many key/value needles are queried across the two turns. Larger variants ask for more values and use longer decode budgets, but do not change the benchmark family. This synthetic family is designed for control: the future-relevant spans are annotated, the second-turn relevance shift is explicit, and every condition shares the same turn-N history. For each variant, we pool balanced, non-recency-favored partitions (three for 4Q, four for 6Q, five for 8Q) whose Q_2 side excludes the tail-recent needles. Appendix Figure 8 summarizes the exact splits.

Evaluation configuration. Per-task base budgets are calibrated to keep each variant in a measurable, non-saturating regime, avoiding both recency saturation and the all-zero baseline regime found at smaller budgets. 2Q has the smallest budget ($B_{\text{base}} = 8192$), so it makes sense that it has the lowest matched baseline. Unless stated otherwise, the fixed sweep is:

- compressor: context-only SnapKV with $R_{\text{ctx}} = 128$ (SnapKV observation-window length);
- base budgets: $B_{\text{base}} = 8192$ for 2Q, $B_{\text{base}} = 16384$ for 4Q, and $B_{\text{base}} = 18432$ for 6Q/8Q;
- restore budgets: $K = 8, 16, 24, 32, 48, 64, 80, 96, 128$;
- replay: one shared full-cache-generated Q_1 transcript per setting and condition.

This removes turn-N error-cascade differences so that the frontier isolates resumed-cache selection quality at turn $N{+}1$. MQ-NIAH-2Q/4Q/6Q use fixed 100-example evaluation sets per partition. MQ-NIAH-8Q is our multi-turn diagnostic variant (§*Repeated relevance shifts*), included in the standard test with 24 examples per partition over five partitions. Prompt and decode settings are reported in the appendix.

Scorers. For turn-N compression, the selector uses the exact generated Q_1 answer tail instead of a generic last-32-token cache suffix. For the main quality curves, we append Q_2 to the compressed active cache, extract the model's post-RoPE Q_2 query projections, and score active and offloaded key rows under those projections before selecting evicted rows for promotion. This exact-Q_2 selector isolates how well the newly revealed turn signal identifies rows worth promoting.

Baselines and references. Every condition reuses the same full-cache-generated turn-N transcript when answering Q_2. We compare the full-cache reference A, the base compressed cache B (the post-Q_1 compressed state before any repair; distinct from the budget B_{base}), matched no-repair B_{match}, Random-K and Oldest-K restore controls, and RepairKV. Random-K and Oldest-K promote from the same evicted-KV store as RepairKV without future-query scoring, so they test whether gains come from query-aware repair or generic reinsertion. The specificity study additionally uses StaleQ-K (score with the previous-turn query), WrongQ-K (score with another example's Q_2), and Refresh-buffered, a stronger Q_2-time reference that reselects the whole $B_{\text{base}} + K$ context budget from active and offloaded rows without full-prefix recomputation. We use Refresh-buffered to expose selector headroom. The main claim is that idle-window repair is useful under matched active cache, not that the current selector is optimal among all Q_2-aware reselectors. Two further time-matched references are used in this version: Refresh-K-budgeted, the unbudgeted Refresh-K with a wall-clock cap matched to RepairKV's measured T_{repair}; and PageSummary-Quest-inspired, a Quest/ShadowKV-style two-stage chunk scorer adapted to the lifecycle slot. Definitions and budget protocol are given in Appendix C.1.

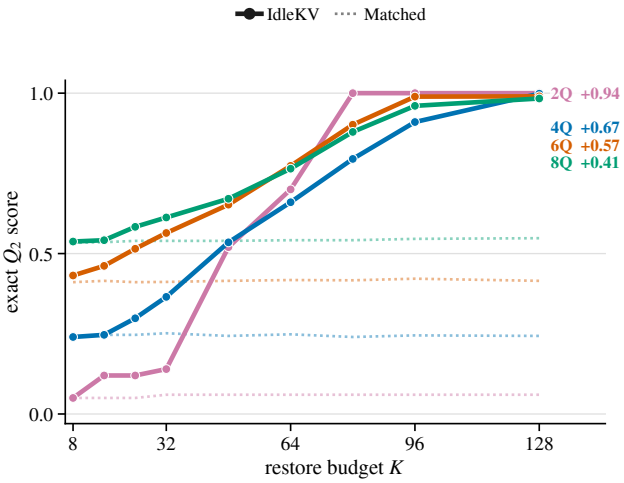

Figure 2. Raw exact Q_2 score under matched resumed-cache budgets. Solid curves are RepairKV. Dotted curves are matched no-repair. Right labels report query count and Δ_{96}, the score gain over matched no-repair at $K = 96$.

5. Results

Matched-budget frontier. Across the 2Q/4Q/6Q/8Q MQ-NIAH family, RepairKV separates from matched no-repair at moderate-to-large restore budgets under the same resumed active-cache budget. At $K = 96$, RepairKV scores 0.917 on 4Q (re-evaluated under the post-fix scorer; Appendix C.1), 0.989 on 6Q, and 0.960 on 8Q, while matched no-repair scores 0.208, 0.422, and 0.546, respectively; Figure 2 reports the full raw Q_2 answer-score frontier. On 4Q, RepairKV exceeds PageSummary-Quest-inspired by $\Delta{=}{+}0.722$ at $K{=}96$ (Holm-corrected $p{<}10^{-9}$, Hodges–Lehmann CI $[0.75, 0.75]$) and tracks the unbudgeted Refresh-K ceiling within paired difference $\Delta{=}{-}0.083$ (TOST-equivalent at margin 0.20); Appendix C.1 reports the per-K contrasts and the per-budget tight-budget supplement. Random-K and Oldest-K stay near matched no-repair in the same evaluations, showing that generic reinsertion from the same offloaded evicted-KV store does not explain the effect. MQ-NIAH-2Q saturates by $K = 80$, while matched no-repair remains near 0.06. Across all twelve balanced partitions at $K{=}128$, RepairKV beats B_{match} and both content-agnostic restore controls; the smallest partition-level gains are 0.585, 0.387, and 0.292 for 4Q, 6Q, and 8Q, respectively (Appendix Figure 8).

Next-turn signal specificity. Repair gains are specific to the newly revealed current-turn signal. At $K = 48$, StaleQ-K and WrongQ-K stay near matched no-repair, with gains 0.028 $[-0.021, 0.083]$ and $[-0.014, 0.069]$, while RepairKV reaches 0.326 $[0.243, 0.410]$. Refresh-buffered's perfect score confirms the right tokens persist in the ac-

Table 1. Next-turn signal specificity at $K = 48$ on MQ-NIAH-4Q (72 paired example-partition cases). Stale and donor controls test whether any query-like signal suffices; Refresh is a full-budget reselector over active and offloaded KV.

Condition	Q_2 score	Δ vs. matched [95% CI]
Matched no-repair	0.215	–
StaleQ-K	0.243	+0.028 [−0.021, 0.083]
WrongQ-K	0.243	+0.028 [−0.014, 0.069]
RepairKV	0.542	+0.326 [0.243, 0.410]
Refresh-buffered	1.000	+0.785 [0.722, 0.847]

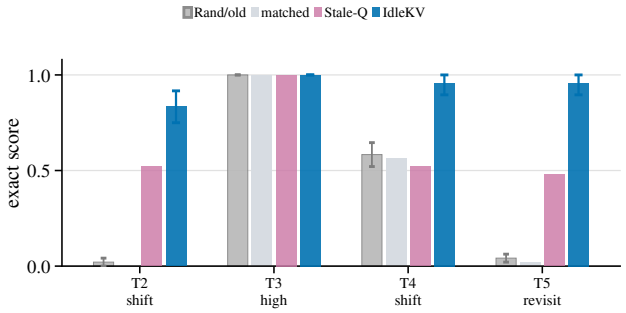

Figure 3. Five-turn relevance-shift diagnostic on MQ-NIAH-8Q (B_{base}=18,432, K=80, n=24). Groups show post-priming exact scores at equal active-cache budget. Intervals are bootstrap 95% CIs for RepairKV, and Stale-Q uses the previous-turn query.

tive and offloaded keys under matched conditions, so idle-window repair is feasible in principle. RepairKV's gain over matched no-repair is preliminary evidence that even a simple K-promotion selector captures meaningful gains, and designing stronger selectors is an open research direction.

Repeated relevance shifts. The repair effect also survives repeated controlled relevance shifts. On a fixed five-turn MQ-NIAH-8Q revisit schedule ($K = 80$, $n = 24$), RepairKV gains 0.542 over matched no-repair on non-initial turns with paired bootstrap interval $[0.458, 0.620]$. Random-K and Oldest-K gain only 0.010 and 0.021. RepairKV remains above StaleQ-K, supporting repeated cue-conditioned adaptation under controlled relevance shifts. End-to-end task validation requires trace-based agent benchmarks.

Eviction-policy sensitivity. The adaptation effect persists under two protocol-matched compression policies beyond the primary SnapKV-style policy: an H_2O-style accumulated-attention retention and a StreamingLLM-style sink-plus-recent rule. The structurally different sink-plus-recent policy (StreamingLLM-style: preserve initial sink tokens plus a recent window (Xiao et al., 2024)) also passes the gate: at $K = 128$, RepairKV gains 0.431 over matched no-repair while Random-K/Oldest-K remain at the matched baseline. Canonical reproductions of H_2O and StreamingLLM would require the full original online policies. A separate Scissorhands-style persistence stress test (Liu et al., 2023) on MQ-NIAH-6Q also passes the locked gate (Appendix Table 3).

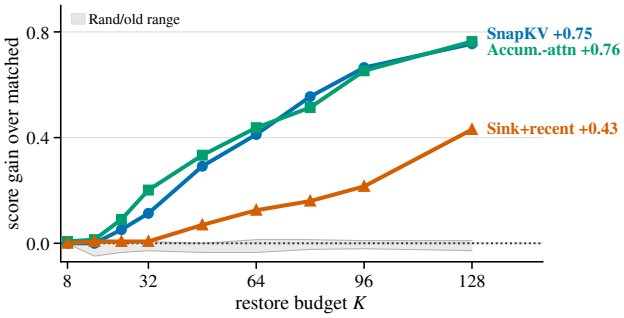

Figure 4. First-stage eviction-policy sensitivity on MQ-NIAH-4Q (n=24, B_{base}=16,384). Curves show exact Q_2 gain over each policy's matched no-repair baseline. The gray band spans Random-K/Oldest-K gains.

Real-repository diagnostic. To test external validity beyond MQ-NIAH, we ran a controlled real-repository relevance-shift diagnostic on 48 callsite examples (locations in source where the model must recover a target identifier such as the called function name) from 12 open-source repositories drawn from the SWE-bench pool (Jimenez et al., 2024) at fixed commits. Each example loads 32K tokens of line-numbered file cards from one repository as the model's context. The first turn Q_1 asks about a file unrelated to the answer, biasing the post-Q_1 compression away from Q_2-relevant content so that the answer-bearing tokens are likely evicted to W_N. The second turn Q_2 is the callsite question. Between turns, an *event* arrives: a description of the callsite with the target identifier redacted. The event serves as the next-turn cue s for repair, and all conditions decode the same event-plus-question prompt (the event followed by a question asking for the target identifier). At $K = 192$, event-only RepairKV (using the event as the only signal) improves exact identifier accuracy from 18.8% matched to 72.9%. Non-query controls and ToolFile-K (file-name-assisted: restricts candidate KV rows to the file named in the event, then backfills with the oldest evicted rows) remain near matched. File-gated RepairKV (label-assisted: restricts candidates to the file named in the event) reaches 83.3%. AnchorWindow-K (oracle locality reference: anchors candidates to the ground-truth code region containing the answer) reaches 89.6%. This remains a small external-validity diagnostic, with no claim to be a real-code benchmark. Strict exclusions (removing examples where the matched cache already retained the answer) keep the gain positive. Appendix Figure 6 gives the full audit.

Runtime. A separate GPU runtime probe measures the repair operation as Q_2 projection plus chunked scan, top-K, and KV movement over pinned host-memory BF16 tensors with Qwen-shaped dimensions. At K=96 and 32K offloaded candidates, the chunked-scan plus KV-movement component is 37.6 ms p95; together with the per-pause Q_2 projection (∼74 ms p95), the full repair operation costs ∼110 ms p95. The component grows to 296 ms at 256K and 1.18 s at 1M; repair cost is approximately constant in the active-cache size and scales with the offloaded candidate pool, which the runtime can bound by trimming the host-memory store. As reference, full-prefix prefill at 32K measures ∼2.13 s p95 with SDPA; FlashAttention-2/3 typically shaves 30–50% off (Dao, 2024; Shah et al., 2024), putting the repair-vs-recompute ratio in the 10×–19× range. The GPU-resident scoring path produces identical quality to the CPU-scored reference within paired difference 0.000 at K=96 (Appendix C.1), so the runtime probe and the matched-budget quality numbers measure the same operator. These figures fit the tool-call gap distributions reported in pause-aware serving (Li et al., 2026).

6. Discussion and Limitations

REPAIRKV is a between-turn maintenance primitive for making KV compression revisable: a runtime adaptation operator over active KV state for resource-adaptive tiered-KV runtimes. It complements existing compression, offload, query-aware access, and pause/resume serving systems (Kwon et al., 2023; Abhyankar et al., 2024; Liu et al., 2025; Dzikanyanga et al., 2026; Li et al., 2026; Zhang et al., 2026) by asking how active memory can be adapted after compression but before resumed decoding. The mechanism should be most useful when:

- the next turn changes which old context matters;
- the compressed cache still leaves useful choices in the evicted host-memory store; and

Table 2. Preliminary real-repository diagnostic at $K = 192$ over 48 callsite cases from 12 open-source repositories (SWE-bench pool, fixed commits). Accuracy is exact target-identifier recovery. Non-query summarizes the best of Random-K, Oldest-K, StaleQ-K, and WrongQ-K analogues. AnchorWindow and File-gated are label-assisted; non-label methods are Matched, RepairKV, and ToolFile-K.

Condition	Accuracy
Matched no-repair	18.8%
Best non-query control	16.7%
ToolFile-K	18.8%
REPAIRKV	72.9%
File-gated REPAIRKV	83.3%
AnchorWindow-K	89.6%

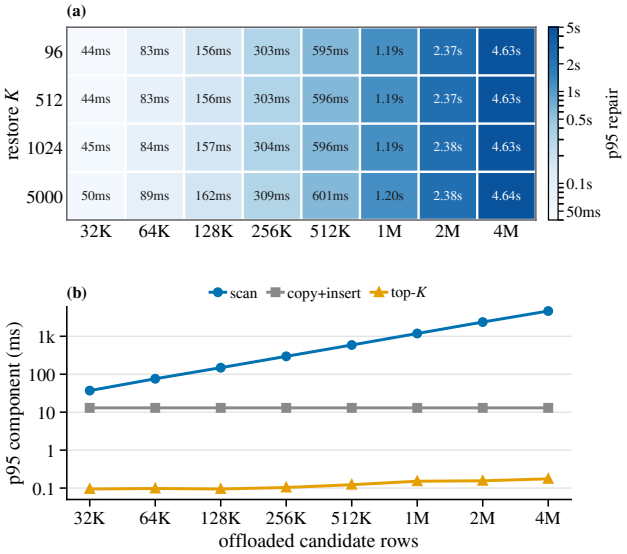

Figure 5. Runtime-capacity probe on the evaluation GPU. (a) Component-summed p95 service time for chunked scan, top-K selection, and KV movement at query length 64 over pinned host-memory BF16 KV-shaped tensors. (b) Stage-wise p95 component time for $K = 5000$. Generation and answer evaluation are excluded.

- the idle window or scheduler has enough slack for scoring and KV movement.

The experiments isolate one paused request on one GPU, matching local or dedicated agent inference but omitting multi-tenant contention. In shared serving, a scheduler would still need to account for HBM blocks, copy bandwidth, and queueing slack. The quality comparisons match active GPU KV budget; total host-plus-device storage and end-to-end service cost are reported separately. Host-memory store size, transfer bytes, Q_2 projection cost, scoring cost, and merge cost are stage-decomposed in Figure 5 and Appendix C.1. PageSummary-Quest-inspired and Refresh-K-budgeted are the time-matched Q_2-aware references in this version; full-prefix recompute and persistent low-rank indices remain orthogonal directions that may further improve the quality-runtime frontier when extra slack or pre-built indices are available. The real-repository diagnostic is small and exploratory, so it supports external validity without serving as a SWE-bench issue-resolution benchmark. Confirmatory experiments use Qwen2.5-7B-Instruct on MQ-NIAH variants at 32K context with SnapKV-style first-stage compression; cross-model evidence is limited to a Llama-3.1-8B-Instruct appendix run. The runtime probe compares to a fresh full-prefix prefill, so deployments with aggressive prefix caching could see different ratios. A non-needle multi-turn evaluation and a multi-model cross-cut are the natural next experiments. The Phase 18 pre-registration chain (plan, scope and gate-logic amendments, the PageSummary-Quest-inspired score-fusion fix, and pre-

rerun outcome bands) is documented in Appendix C.1.

Conclusion. REPAIRKV makes the case for revisable KV compression. If context relevance changes after eviction, the cache lifecycle can move in both directions: compress to fit the active budget, then repair by promoting relevant evicted state before the next decode. Across MQ-NIAH-2Q/4Q/6Q/8Q, this repair beats matched no-repair while random and oldest-token promotions stay near baseline. The broader direction is mutable KV memory for resource-adaptive long-context inference, with pause windows treated as part of the inference budget.

Future work. Our work highlights the need for a production-facing repair benchmark that keeps the matched-budget requirements used here, but adds the features needed to evaluate deployment relevance: multi-turn traces with explicit pause boundaries, hidden future relevance, shared pre-turn history, and controls that separate active-cache budget from total storage, transfer, and recompute cost. It should then test repair across top open and proprietary long-context models, multiple compression and offload policies, and production-style traces rather than a single synthetic task family. On the systems side, the benchmark should measure the idle-window slack available in real agent workloads such as coding, tool use, retrieval, and browser interaction, since repair is only useful when scoring and KV movement fit the scheduler's time and memory constraints. Finally, preliminary nonlinear variable-tracking diagnostics (Appendix B) suggest that compression may fail especially sharply when relevant information forms branching or conflicting dependency graphs rather than linear chains. Turning that observation into a robust benchmark is a central next step; REPAIRKV appears robust in preliminary tests, which would further establish the need for a dual compression-repair mechanism, but extensive broader validation is still needed.

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A. Additional Evaluation Details

Protocol details. Concise split prompts request values only, comma-separated, with a short answer prefix and fixed decode budgets of 24 tokens for MQ-NIAH-4Q, 48 tokens for MQ-NIAH-6Q, and 64 tokens for MQ-NIAH-8Q. Generation is deterministic greedy decoding with no sampling. The main examples use dataset seed offset 0. Random-K uses a deterministic example- and K-dependent seed. Bootstrap intervals are percentile intervals over paired example-partition rows with fixed seeds. Main Qwen runs use BF16 weights on one RTX PRO 6000 Blackwell GPU with Python 3.12.3, PyTorch 2.10.0+cu128, and Transformers 5.2.0. The exact-Q_2 attention-query selector appends Q_2 to the compressed active cache, extracts the model's post-RoPE query projections for the Q_2 prompt tokens, and scores active and offloaded key rows under those projections before selecting evicted rows for promotion. This selector is distinct from decode-time answer-token attention and from the final answer metric. The Q_2-appended-state proxy appends Q_2 to the compressed cache and uses the appended cache rows as a second Q_2-conditioned scoring signal.

Span-group diagnostic. SpanRef-K starts from the same B state and offloaded evicted-KV store as RepairKV. It enumerates the small family of annotated Q_2 answer-span-group subsets, one key/value span group per requested needle, evaluates each subset once under actual Q_2 generation, and chooses the highest-scoring subset with total cost at most K. It is exact over this benchmark-specific span-group family, but not over all arbitrary token subsets or possible final generations. Because the selected span-group subset may use fewer than K rows, a repair selector can occasionally score higher at low budgets by spending the full budget on useful local neighborhoods around future-relevant spans. Thus SpanRef-K is a diagnostic span-group reference, not a ceiling.

Partition choices. For MQ-NIAH-4Q, the pooled panel uses $14{\to}23$, $24{\to}13$, and $34{\to}12$. These are the balanced $2|2$ partitions whose Q_2 side excludes the tail-anchored fourth needle. For MQ-NIAH-6Q, the pooled panel uses $156{\to}234$, $256{\to}134$, $356{\to}124$, and $456{\to}123$, so the second turn excludes the two latest needles. For MQ-NIAH-8Q, the pooled panel uses $5678{\to}1234$, $1678{\to}2345$, $2678{\to}1345$, $3678{\to}1245$, and $4678{\to}1235$, again excluding the two latest needles from the second turn. Within-turn order is not treated as a separate condition because the prompt asks for comma-separated values only and scoring checks whether each requested value is recovered. Recency-favorable splits such as $12{\to}34$ are left to separate diagnostics outside the main future-query repair result.

Scissorhands-style persistence stress test. Table 3 repeats the MQ-NIAH-6Q locked protocol after replacing the initial SnapKV-style eviction policy with a one-shot persistence policy inspired by Scissorhands. The test asks whether repair still works when the compressed active cache is chosen by past attention persistence in place of the primary eviction policy. Canonical online per-head Scissorhands reproduction remains future work.

Table 3. Scissorhands-style persistence stress test on MQ-NIAH-6Q (B_{base}=18,432, n=24). Values are exact scores. Best ctrl. is the stronger of Random-K and Oldest-K; the interval is the paired bootstrap 95% CI for RepairKV$-B_{\text{match}}$.

K	Full	B_{match}	Best ctrl.	RepairKV	Gain CI
48	0.990	0.365	0.382	0.712	[0.292, 0.403]
64	0.990	0.365	0.382	0.795	[0.378, 0.490]
80	0.990	0.378	0.385	0.934	[0.503, 0.604]
96	0.990	0.375	0.382	0.993	[0.566, 0.670]
128	0.990	0.368	0.378	0.990	[0.573, 0.674]

B. Additional Discussion

Tiered-cache scaling. Repair creates a two-level cache-budget problem because the main curves match the resumed active GPU KV budget while tracking total host-plus-device storage separately. A strict device-resident policy chooses the active-cache budget, while a looser off-device policy decides which evicted rows remain searchable, compressed, summarized, or recomputable. In the Qwen-shaped accounting above, 4M searchable rows already imply about 240 GB before metadata. Thus RepairKV supports the middle operation in a tiered hierarchy: turn-conditioned promotion from host memory during an idle window. This is a hardware-facing implication; hardware measurement remains future work.

Idle-window repair interface. A practical implementation would expose a host-memory index over evicted KV units, a time-budgeted selector conditioned on the next-turn signal, asynchronous promotion into the active cache, and fallbacks to

no-repair or full recompute. This turns a pause into a schedulable resource: the system can spend only the slack that remains after queueing, transfer, and generation constraints, and can move the selector from token rows to pages or compressed metadata as hardware and serving constraints require.

Preliminary nonlinear variable-tracking diagnostic. As an exploratory stress test, we also tested a modified RULER variable-tracking task that makes the dependency structure less linear than the standard 4-hop chain. The diagnostic uses an 8-hop permuted chain with two distractor divergences: some assignments introduce branches that are not used by the final answer, so the model must recover the relevant path through a small information graph rather than follow a tail-friendly linear sequence. In 32K-context Qwen2.5-7B-Instruct runs, SnapKV compression degraded this task sharply even at moderate retained-cache budgets. Preliminary repair runs recovered substantial accuracy from the same compressed states using small restore fractions. These results are not part of the main evidence, but they motivate a broader benchmark for nonlinear and conflicting relevance patterns, where eviction policies may fail even when standard long-context tasks appear solved.

Future benchmark axes. Future benchmarks should vary the number, timing, and conflict structure of new-turn constraints. The 2Q and 8Q breadth checks suggest that easy regimes can saturate while stricter multi-query turns expose how quickly eviction failures accumulate. The five-turn diagnostic strengthens the dynamic-workflow evidence, but it remains synthetic. Broader task families should keep the same matched-budget controls while adding real tool traces and stronger selector families.

C. Supplementary Experimental Views

The supplementary figures probe external validity, query-count and operating-regime breadth, partition consistency, runtime capacity, and appendix selector diagnostics. Exact audit values remain in released result artifacts.

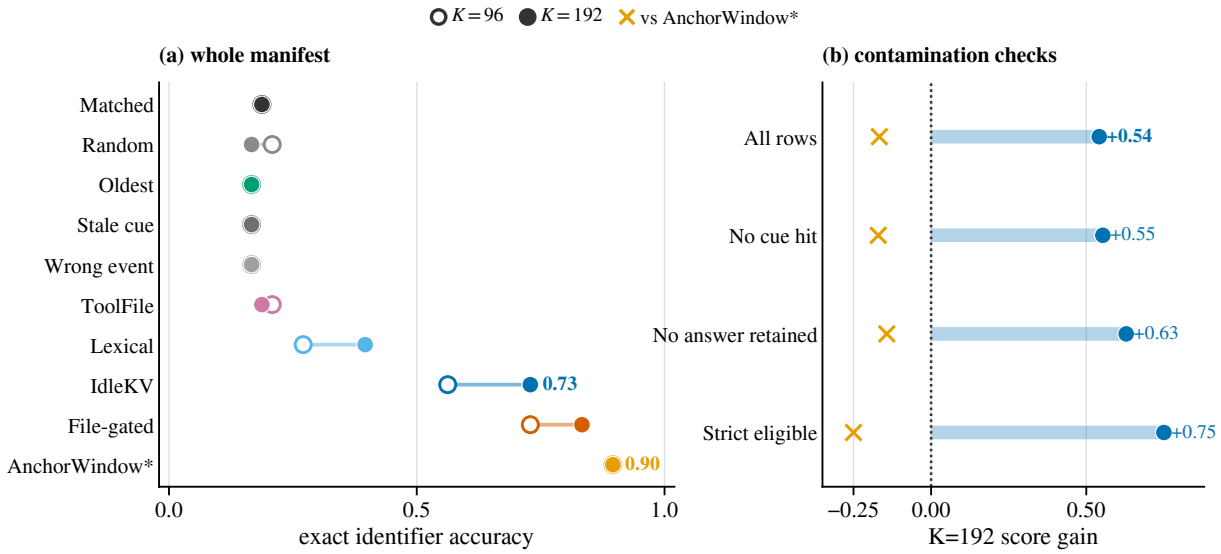
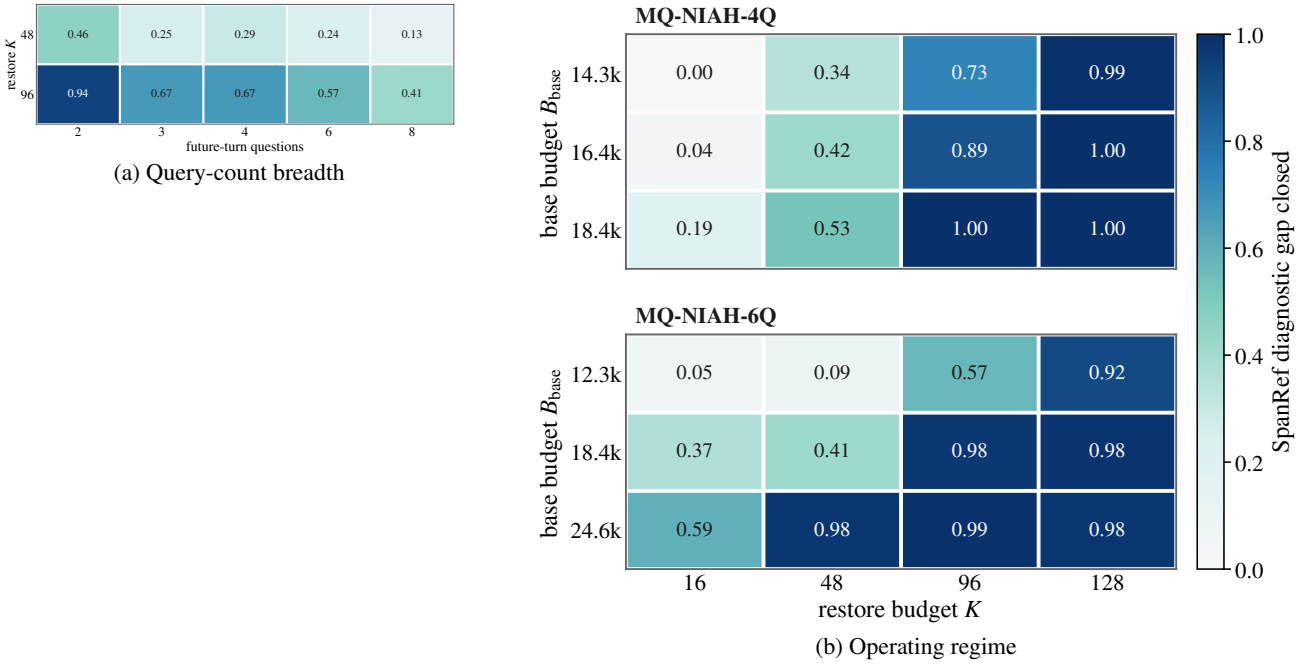


Figure 6. Controlled real-repository relevance-shift diagnostic over 48 callsite examples from 12 repositories drawn from the SWE-bench repository pool at pinned commits. Repair sees only the event cue. All conditions decode the same event-plus-question prompt. Left: exact identifier accuracy at $K = 96$ and $K = 192$. Right: $K = 192$ REPAIRKV gain over matched no-repair after removing cue-only or answer-retention rows. ToolFile-K is file-name-assisted with oldest-row backfill. File-gated REPAIRKV restricts candidates to the event-named file before applying the event-only scorer. Lexical is an answer-sanitized diagnostic outside the deployable-selector comparison. AnchorWindow-K is a label-assisted locality reference outside the deployable-baseline and end-to-end issue-resolution comparisons.



(a) Query-count breadth

(b) Operating regime

Figure 7. Supplementary breadth diagnostics. (a) Cells show RepairKV score gain over matched no-repair for $K = 48$ and $K = 96$ across 2-, 3-, 4-, 6-, and 8-question split-query variants. (b) Cells show the fraction of available SpanRef-K diagnostic gain recovered by RepairKV, clipped to $[0, 1]$, for MQ-NIAH-4Q/6Q. Saturated cells with little matched-to-SpanRef-K gap should not be read as upper-bound statements.

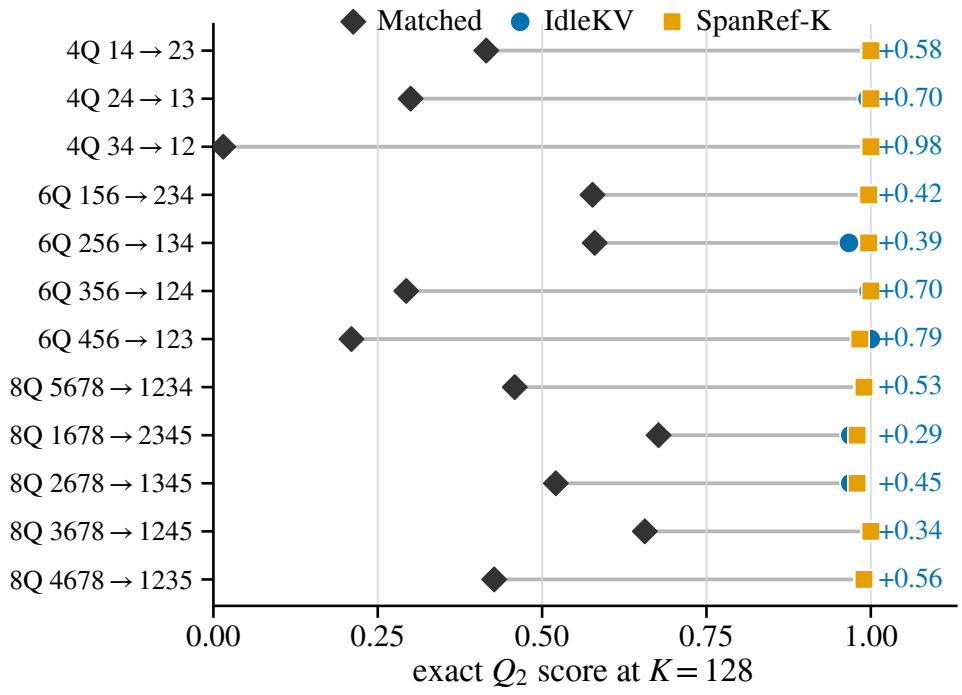


Figure 8. Per-partition endpoint scores at $K = 128$ on the final exact evaluations. Gray segments show the change from matched no-repair to RepairKV, orange points show the benchmark-metadata SpanRef-K diagnostic, and labels report the RepairKV difference over matched no-repair.

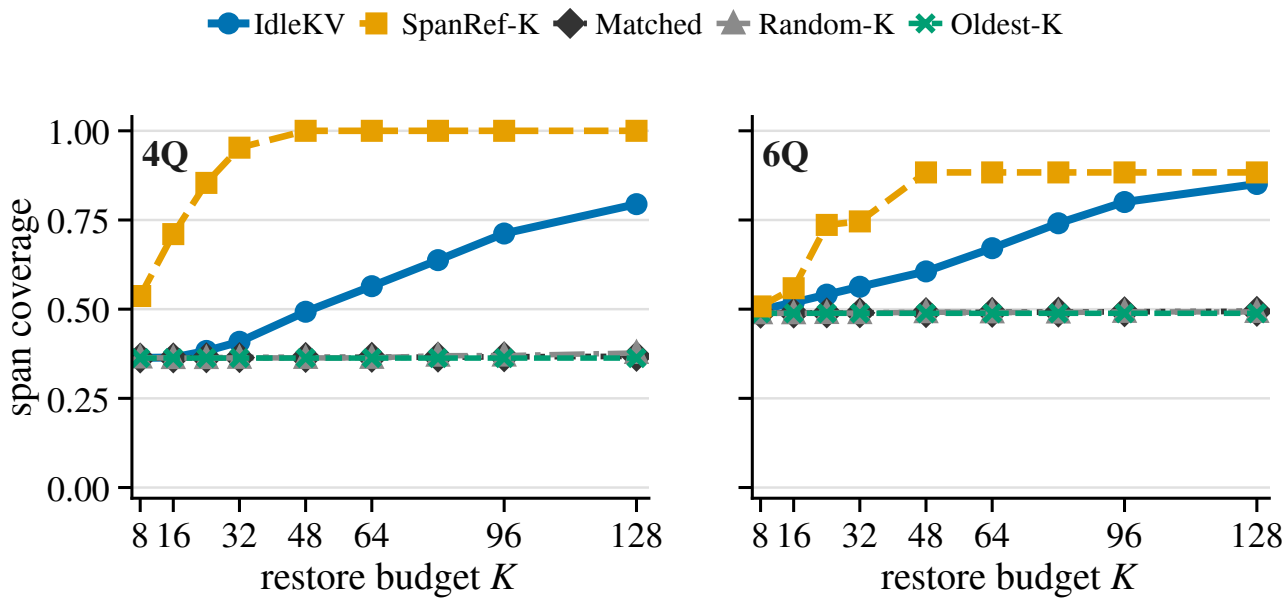
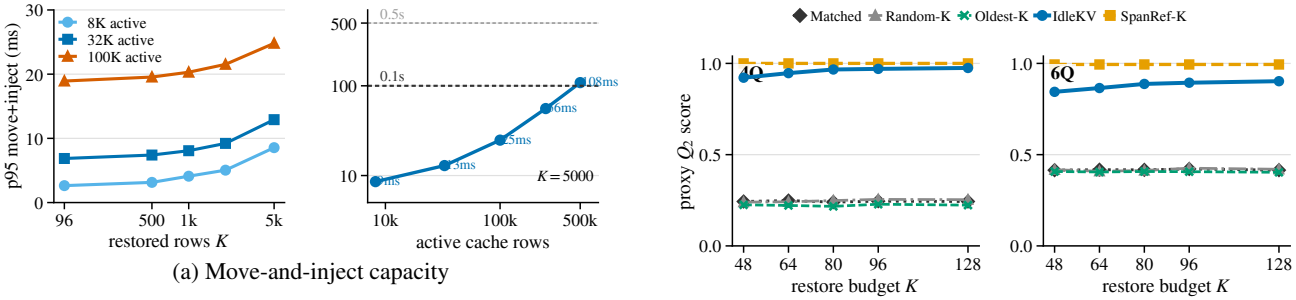


Figure 9. Mechanism diagnostic for the final exact MQ-NIAH evaluations. Curves show overlap between promoted tokens and annotated Q_2 answer spans. Content-agnostic promotions stay near baseline while REPAIRKV increasingly covers future-relevant spans.



(a) Move-and-inject capacity

(b) Proxy-selector diagnostic

Figure 10. Supplementary runtime and selector diagnostics. (a) Synthetic move-and-inject capacity for Qwen-shaped BF16 KV on the evaluation GPU, separate from the Figure 5 scan/select envelope. Measurements use pinned host-memory promoted blocks and report p95 latency as K or active cache size grows. (b) Appendix proxy-selector diagnostic on MQ-NIAH-4Q/6Q (100 examples per split). At $K = 96$, content-agnostic promotions stay within 0.010/0.004 of matched no-repair while proxy REPAIRKV remains well above matched.

C.1. Time-matched evaluation supplement

Time-matched baselines. Refresh-K-budgeted reuses the unbudgeted Refresh-K scorer with a wall-clock cap matched to REPAIRKV's measured per-example T_{repair} (selection + inject + transfer + amortized Q_2 projection and scoring); the chunk-restricted softmax denominator is (active ∪ chunk-evicted) per chunk, which the cap forces because a global denominator across all evicted positions is incompatible with a wall-clock budget. PageSummary-Quest-inspired computes per-chunk max-key summaries at compression time, ranks chunks by a cheap Q_2-summary inner product, and within budget visits top-ranked chunks with the same per-position scorer used by Refresh-K-budgeted; Stage-1 ranking serves as the fallback when the Stage-2 cap fires before any chunk is fully scored. Both baselines are matched to REPAIRKV on resumed active-cache rows ($B_{\text{base}} + K$).

Wall-clock-matched K-sweep. Table 4 reports MQ-NIAH-4Q at a 150 ms absolute scoring budget across $K \in \{32, 64, 96, 128\}$ on Qwen2.5-7B-Instruct (n=36 paired observations per K). At this budget the Refresh-K-budgeted cap

fires for all $36/36$ examples, scoring approximately $4{,}000/32{,}768$ evicted positions per call; REPAIRKV dominates at $K \geq 64$.

Table 4. Time-matched K-sweep at 150 ms absolute scoring budget on MQ-NIAH-4Q (Qwen2.5-7B, B_{base}=16384, n=36). RKB and PSum apply the budget to their scoring loops; REPAIRKV and Refresh-K (unbudgeted) are reported for context.

K	A	B_{match}	REPAIRKV	Refresh-K	RKB (150 ms)	PSum (150 ms)
32	1.000	0.208	0.375	1.000	0.500	0.208
64	1.000	0.208	0.639	1.000	0.486	0.208
96	1.000	0.208	0.917	1.000	0.472	0.194
128	1.000	0.181	1.000	1.000	0.514	0.194

Lifecycle-slot vs. burst attribution. At K=96 on Qwen, REPAIRKV-no-burst (the lifecycle-slot scorer without burst expansion) reaches 0.653, exceeding PageSummary-Quest-inspired by Δ_{slot}=+0.459; full REPAIRKV reaches 0.917, exceeding the no-burst variant by Δ_{burst}=+0.264. Both contributions are material, with the lifecycle-slot share larger.

GPU–CPU scoring equivalence. Activating the GPU-resident path for SCORE (matmul and softmax kept on the model device) reproduces the CPU-scored quality numbers exactly: at K=96 on Qwen, paired difference is 0.000 across n=36 for A, B_{match}, and REPAIRKV. The W2 runtime probe and the W1 quality numbers therefore measure the same operator.

Pre-registration chain. The Phase 18 plan was committed to git before any GPU run touched headline numbers (commit `601d807`). A scope amendment (`55e8bda`, pre-data) and a gate-logic correction (`af2fd93`, post-data, replacing a $\Delta \geq 0.10$ threshold against Refresh-K-budgeted with a TOST-equivalence criterion that matches the abstract's "approaches" wording) followed; we report both verdicts in the released audit. After the K-sweep, an implementation bug in the PageSummary-Quest-inspired score-fusion (`c1f08a7`) was identified and fixed; outcome bands for the rerun were committed before any rerun read its data (`e437c19`). The K-sweep was re-run under the corrected scorer.